UC SAN DIEGO / ECE 251C

11
Wavelets from filters

Cascade iteration, infinite products, and the scaling-function construction

How can a few filter taps determine a scaling function?

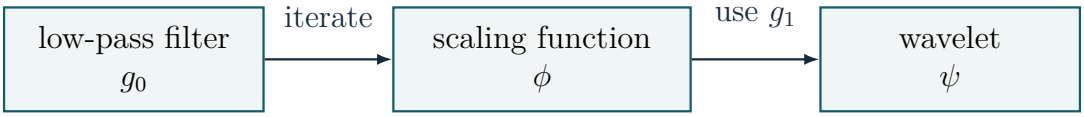
11.1 Reversing the construction

So far, scaling functions and wavelets have supplied the filters for a discrete wavelet transform. We now reverse that direction. Given the filters, can we recover the continuous-domain functions that they represent?

Recall the two-scale equations:

$$\begin{aligned} \phi(t) &= \sqrt{2} \sum_{n \in \mathbb{Z}} g_0[n] \phi(2t - n), \\ \psi(t) &= \sqrt{2} \sum_{n \in \mathbb{Z}} g_1[n] \phi(2t - n). \end{aligned} \tag{11.1}$$

The first equation is implicit: the unknown ϕ occurs on both sides. Once ϕ is known, the second equation constructs ψ directly.



convergence is the new issue

Figure 11.1: Filters determine functions through refinement. A valid finite-stage filter bank is not, by itself, a convergence theorem for infinitely many stages.

Choose an integrable, finite-energy initial function $\phi^{(0)}$ and repeatedly apply the same refinement rule:

$$\boxed{\phi^{(k+1)}(t) = \sqrt{2} \sum_{n \in \mathbb{Z}} g_0[n] \phi^{(k)}(2t - n).} \tag{11.2}$$

This is the *cascade algorithm*. The superscript (k) counts iterations, not derivatives. Throughout this lecture the filters are FIR, so each refinement sum is finite. We retain g_0 for the refinement filter and $h_0[n] = g_0^*[-n]$ for the analysis filter. The symbol j is reserved for the imaginary unit; i and I will index products and their truncation levels.

11.2 Fixed points and the normalization of the iterates

A *fixed point* of a map F is an object x satisfying $F(x) = x$. If $x_{k+1} = F(x_k)$ converges to x and F is continuous there, then

$$F(x) = F\Big(\lim_{k\to\infty} x_k\Big) = \lim_{k\to\infty} F(x_k) = \lim_{k\to\infty} x_{k+1} = x.$$

Continuity is what permits the limit to pass through F.

A scalar example. If $x_{k+1} = x_k/2$, then

$$x_1 = \frac{x_0}{2}, \quad x_2 = \frac{x_0}{4}, \quad x_k = \frac{x_0}{2^k} \longrightarrow 0.$$

The limit is the fixed point $0 = 0/2$. For a scaling function, we seek a nonzero fixed point of the operator

$$(\mathrm{C}_{g_0} u)(t) := \sqrt{2} \sum_n g_0[n] u(2t - n).$$

This operator is continuous in L^2. Indeed, changing variables gives

$$\|\sqrt{2}\, u(2\,\cdot - n)\|_{L^2}^2 = 2 \int_{\mathbb{R}} |u(2t - n)|^2 \,\mathrm{d}t = \int_{\mathbb{R}} |u(s)|^2 \,\mathrm{d}s.$$

The triangle inequality therefore implies

$$\|\mathrm{C}_{g_0} u - \mathrm{C}_{g_0} v\|_{L^2} \le \left(\sum_n |g_0[n]|\right) \|u - v\|_{L^2}.$$

Thus, if $\|\phi^{(k)} - \phi\|_{L^2} \to 0$, continuity gives $\mathrm{C}_{g_0}\phi = \phi$. This does not yet prove that the iterates converge.

Preserving the integral. For an integrable u, the substitution $s = 2t - n$ gives

$$\int_{\mathbb{R}} \mathrm{C}_{g_0} u(t) \,\mathrm{d}t = \frac{1}{\sqrt{2}} \left(\sum_n g_0[n]\right) \int_{\mathbb{R}} u(t) \,\mathrm{d}t.$$

Hence the normalization $\sum_n g_0[n] = \sqrt{2}$ from Lecture 10 preserves the integral at every iteration. We start with integral one. This fixes the otherwise arbitrary amplitude of a nonzero refinement solution.

11.3 A three-tap cascade: from a box toward a tent

Consider the three-tap refinement filter

$$g_0[0] = \frac{1}{2\sqrt{2}}, \qquad g_0[1] = \frac{1}{\sqrt{2}}, \qquad g_0[2] = \frac{1}{2\sqrt{2}}, \tag{11.3}$$

with all other taps zero. Its sum is $\sqrt{2}$. The cascade becomes

$$\phi^{(k+1)}(t) = \frac{1}{2}\phi^{(k)}(2t) + \phi^{(k)}(2t-1) + \frac{1}{2}\phi^{(k)}(2t-2). \tag{11.4}$$

Start with $\phi^{(0)} = \mathbf{1}_{[0,1)}$. Each new iterate is a sum of three compressed and shifted copies of the previous one.

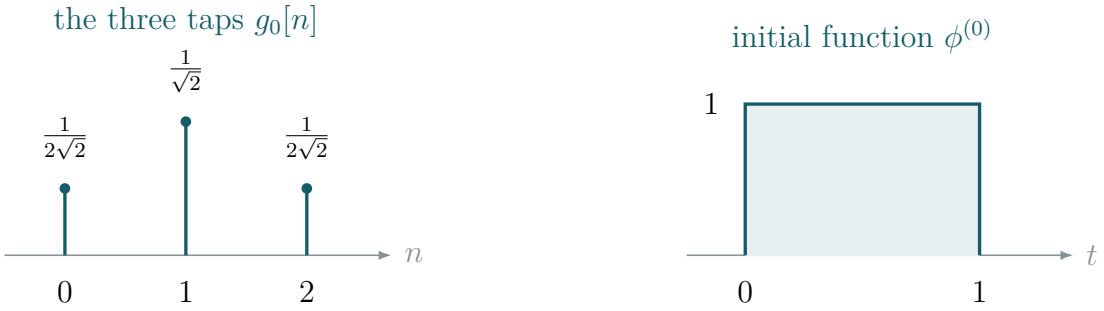


Figure 11.2: The weights in the refinement equation are $\sqrt{2}g_0 = (1/2, 1, 1/2)$. The initial box has integral one.

For the first iteration, the three supports are $[0, 1/2)$, $[1/2, 1)$, and $[1, 3/2)$. They do not overlap, so

$$\phi^{(1)}(t) = \begin{cases} \frac{1}{2}, & 0 \le t < \frac{1}{2}, \\ 1, & \frac{1}{2} \le t < 1, \\ \frac{1}{2}, & 1 \le t < \frac{3}{2}, \\ 0, & \text{otherwise.} \end{cases} \tag{11.5}$$

Its integral remains one:

$$\frac{1}{2}\cdot\frac{1}{2} + 1\cdot\frac{1}{2} + \frac{1}{2}\cdot\frac{1}{2} = 1.$$

The filter controls both the weights and the relative positions of the copies. The next iteration uses this new staircase, not the original box.

The second iteration: add the overlaps. Each term in (11.4) now has cells of width $1/4$. On $[m/4, (m+1)/4)$, for $m = 0, \dots, 6$, the resulting heights are

$$\frac{1}{4}, \quad \frac{1}{2}, \quad \frac{3}{4}, \quad 1, \quad \frac{3}{4}, \quad \frac{1}{2}, \quad \frac{1}{4}.$$

For example, on $1/2 \le t < 3/4$,

$$\phi^{(1)}(2t) = \frac{1}{2}, \qquad \phi^{(1)}(2t-1) = \frac{1}{2}, \qquad \phi^{(1)}(2t-2) = 0.$$

Thus $\phi^{(2)}(t) = \frac{1}{2}(\frac{1}{2}) + \frac{1}{2} + \frac{1}{2}(0) = \frac{3}{4}$ on that cell. The other cells follow by the same substitution.

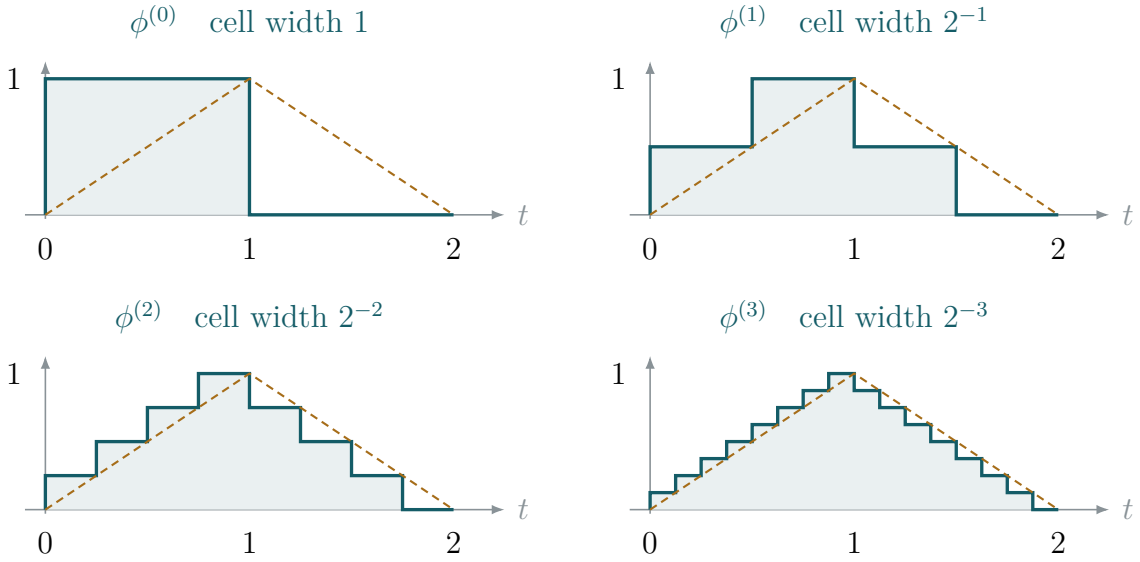


Figure 11.3: Four cascade iterates on the same axes. The dashed tent is the candidate limit. The support approaches $[0, 2]$ while the cell width halves at each step.

One more iteration. The cells now have width $1/8$. From left to right, their nonzero heights are

$$\frac{1}{8}(1, 2, 3, 4, 5, 6, 7, 8, 7, 6, 5, 4, 3, 2, 1).$$

The staircase becomes finer and approaches a tent supported on $[0, 2]$. A picture suggests this limit, but we should check both the proposed fixed point and the convergence of the iterates.

11.4 Checking the tent and its role in the construction

Define

$$\tau(t) = \begin{cases} t, & 0 \le t \le 1, \\ 2 - t, & 1 < t \le 2, \\ 0, & \text{otherwise.} \end{cases} \tag{11.6}$$

We verify that $\tau(t) = \frac{1}{2}\tau(2t) + \tau(2t-1) + \frac{1}{2}\tau(2t-2)$. Split its support at the breakpoints of the three terms:

t	$\frac{1}{2}\tau(2t)$	$\tau(2t-1)$	$\frac{1}{2}\tau(2t-2)$	sum
$[0, \frac{1}{2}]$	t	0	0	t
$[\frac{1}{2}, 1]$	$1-t$	$2t-1$	0	t
$[1, \frac{3}{2}]$	0	$3-2t$	$t-1$	$2-t$
$[\frac{3}{2}, 2]$	0	0	$2-t$	$2-t$

Outside $[0, 2]$, every term is zero. The formulas agree at their endpoints, so the refinement identity holds everywhere.

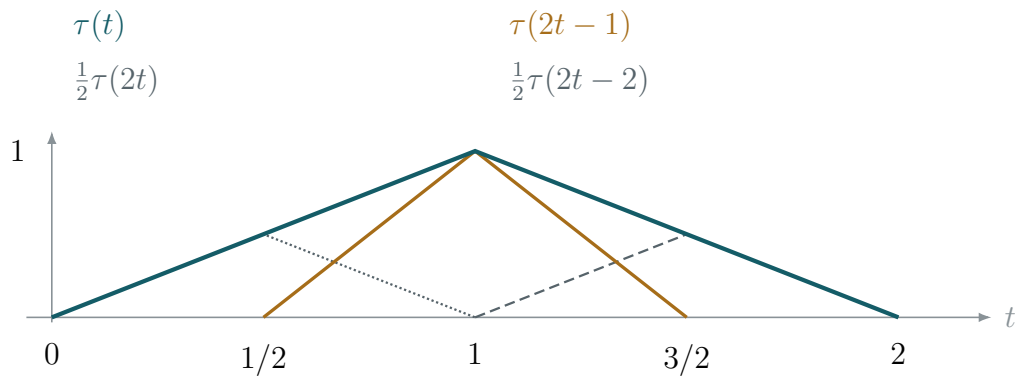


Figure 11.4: The three smaller tents add to the large tent. Where two components overlap, their slopes combine to give the slopes 1 and -1 of τ.

Refinability is not the same as orthogonality. This example illustrates the cascade, but its filter is not an orthogonal scaling filter:

$$\sum_n |g_0[n]|^2 = \frac{1}{8} + \frac{1}{2} + \frac{1}{8} = \frac{3}{4} \neq 1.$$

Adjacent translates of the tent also have a nonzero inner product:

$$\langle \tau, \tau(\cdot - 1) \rangle_{L^2} = \int_1^2 (2-t)(t-1)\,\mathrm{d}t = \int_0^1 (1-u)u\,\mathrm{d}u = \frac{1}{6}.$$

Why these particular iterates converge. Let $M = 2^k$ and write the kth iterate as

$$\phi^{(k)}(t) = \sum_m c_k[m]\mathbf{1}_{[m/M,(m+1)/M)}(t).$$

Initially $c_0[0] = 1$. Substitution in (11.4) gives

$$c_{k+1}[m] = \tfrac{1}{2}c_k[m] + c_k[m - 2^k] + \tfrac{1}{2}c_k[m - 2^{k+1}].$$

With $C_k(z) = \sum_m c_k[m]z^{-m}$, this reads

$$C_{k+1}(z) = \tfrac{1}{2}(1 + z^{-2^k})^2 C_k(z), \qquad C_0(z) = 1.$$

Iterating, and setting $q = z^{-1}$, yields

$$C_k(z) = 2^{-k}\left[\prod_{r=0}^{k-1}(1 + q^{2^r})\right]^2.$$

Repeated difference of squares gives

$$(1 - q)\prod_{r=0}^{k-1}(1 + q^{2^r}) = 1 - q^{2^k}.$$

Comparing with $(1 - q)\sum_{r=0}^{M-1} q^r = 1 - q^M$ proves the polynomial identity

$$C_k(z) = \frac{1}{M}\left(\sum_{r=0}^{M-1} z^{-r}\right)^2.$$

The coefficient of z^{-m} counts pairs of integers in $\{0, \dots, M - 1\}$ whose sum is m. There are $m+1$ such pairs for $0 \le m < M$, and $2M-1-m$ for $M \le m \le 2M-2$. Thus

$$c_k[m] = \begin{cases} (m + 1)/M, & 0 \le m < M, \\ (2M - 1 - m)/M, & M \le m \le 2M - 2, \\ 0, & \text{otherwise.} \end{cases}$$

On every cell meeting $[0, 2]$, including the final zero-height cell, this height equals the tent at the right endpoint. Since the tent changes by at most the cell width,

$$|\phi^{(k)}(t) - \tau(t)| \le 2^{-k}, \qquad \|\phi^{(k)} - \tau\|_{L^2}^2 \le 2\,4^{-k} \longrightarrow 0.$$

The second bound follows by integrating over $[0, 2]$, outside which both functions vanish. This proves convergence, not just the fixed-point identity.

11.5 Fourier analysis turns refinement into multiplication

The cascade combines shifted, compressed functions. Fourier analysis separates those operations into a phase factor and a change of frequency. Assume first that ϕ is integrable, and recall

$$\widehat{\phi}(\omega) = \int_{\mathbb{R}} \phi(t) \mathrm{e}^{-\mathrm{j}\omega t}\, \mathrm{d}t, \qquad G_0(z) = \sum_n g_0[n] z^{-n}.$$

The first expression transforms a function on $\mathbb{R}$; evaluating the second on $z = \mathrm{e}^{\mathrm{j}\theta}$ gives the DTFT of a sequence.

Transform the refinement equation term by term:

$$\widehat{\phi}(\omega) = \sqrt{2} \sum_n g_0[n] \int_{-\infty}^{\infty} \phi(2t - n) \mathrm{e}^{-\mathrm{j}\omega t}\, \mathrm{d}t.$$

For each fixed n, substitute

$$u = 2t - n, \qquad t = \frac{u + n}{2}, \qquad \mathrm{d}t = \frac{1}{2}\, \mathrm{d}u.$$

The limits remain $-\infty$ and $+\infty$, because the slope is positive. Hence

$$\begin{aligned} \widehat{\phi}(\omega) &= \frac{1}{\sqrt{2}} \sum_n g_0[n] \int_{-\infty}^{\infty} \phi(u) \mathrm{e}^{-\mathrm{j}\omega(u+n)/2}\, \mathrm{d}u \\ &= \frac{1}{\sqrt{2}} \left(\sum_n g_0[n] \mathrm{e}^{-\mathrm{j}(\omega/2)n} \right) \left(\int_{\mathbb{R}} \phi(u) \mathrm{e}^{-\mathrm{j}(\omega/2)u}\, \mathrm{d}u \right) \\ &= \frac{G_0(\mathrm{e}^{\mathrm{j}\omega/2})}{\sqrt{2}}\, \widehat{\phi}(\omega/2). \end{aligned}$$

Define the normalized response

$$m_0(\theta) := \frac{G_0(\mathrm{e}^{\mathrm{j}\theta})}{\sqrt{2}}. \tag{11.7}$$

Then

$$\boxed{\widehat{\phi}(\omega) = m_0(\omega/2) \widehat{\phi}(\omega/2).} \tag{11.8}$$

The argument of m_0 is $\omega/2$, not ω. The same compression that replaces t by $2t$ replaces the Fourier argument by $\omega/2$.

11.6 The infinite-product formula

Apply (11.8) repeatedly:

$$\begin{aligned}\widehat{\phi}(\omega) &= m_0(\omega/2)\widehat{\phi}(\omega/2)\\ &= m_0(\omega/2)m_0(\omega/4)\widehat{\phi}(\omega/4)\\ &= m_0(\omega/2)m_0(\omega/4)m_0(\omega/8)\widehat{\phi}(\omega/8).\end{aligned}$$

For a positive integer I, put

$$P_I(\omega) := \prod_{i=1}^{I} m_0(\omega/2^i).$$

The exact finite identity is

$$\widehat{\phi}(\omega) = P_I(\omega)\widehat{\phi}(\omega/2^I). \tag{11.9}$$

Normalize $\int_{\mathbb{R}} \phi(t)\,\mathrm{d}t = \widehat{\phi}(0) = 1$. The Fourier transform of an integrable function is continuous, so for fixed ω,

$$\widehat{\phi}(\omega/2^I) \longrightarrow 1.$$

In particular, this factor is nonzero for all sufficiently large I. Dividing (11.9) by it and taking the limit gives

$$\boxed{\widehat{\phi}(\omega) = \prod_{i=1}^{\infty} \frac{G_0(\mathrm{e}^{\mathrm{j}\omega/2^i})}{\sqrt{2}}.} \tag{11.10}$$

Thus a normalized integrable refinement solution, when it exists, is determined by the filter.

The actual cascade retains its initial function. Applying the same Fourier calculation to (11.2) gives

$$\widehat{\phi}^{(I)}(\omega) = P_I(\omega)\widehat{\phi}^{(0)}(\omega/2^I). \tag{11.11}$$

The factor on the right is the transform of the seed. If its integral is one, that factor tends to one at every fixed frequency. Nevertheless, pointwise convergence of the products is not by itself L^2 convergence of the cascade. By Plancherel,

$$\|\phi^{(I)} - \phi\|_{L^2}^2 = \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{\phi}^{(I)}(\omega) - \widehat{\phi}(\omega)|^2\,\mathrm{d}\omega.$$

An infinite-scale construction must control this full energy. Convergence also depends on the initial function, not only on the filter.

11.7 The fundamental theorem of wavelet analysis

The product formula describes a candidate scaling function. The next theorem gives conditions under which that candidate really generates an orthonormal multiresolution analysis. It separates necessary filter identities from a sufficient construction.

Theorem 11.1 (Mallat–Meyer: necessary conditions and a sufficient construction)**.** Suppose an integrable scaling function ϕ generates an orthonormal MRA, has $\int_{\mathbb{R}} \phi(t)\,\mathrm{d}t = 1$, and has an FIR refinement filter g_0. Then

$$|G_0(\mathrm{e}^{\mathrm{j}\omega})|^2 + |G_0(\mathrm{e}^{\mathrm{j}(\omega+\pi)})|^2 = 2 \quad (\omega \in \mathbb{R}), \tag{11.12}$$

$$G_0(1) = \sum_n g_0[n] = \sqrt{2}. \tag{11.13}$$

Conversely, let an FIR filter g_0 satisfy these two conditions. If, in addition,

$$\boxed{\min_{|\omega|\le \pi/2} |G_0(\mathrm{e}^{\mathrm{j}\omega})| > 0,} \tag{11.14}$$

then the infinite product in (11.10) is the Fourier transform of a compactly supported scaling function $\phi \in L^2(\mathbb{R})$ with integral one. Its integer translates are orthonormal, and the closed spaces

$$V_i = \overline{\operatorname{span}\{2^{i/2}\phi(2^i\,\cdot - n) : n \in \mathbb{Z}\}}^{L^2}, \qquad i \in \mathbb{Z},$$

form an orthonormal multiresolution analysis: they are nested, satisfy the scaling and translation rules, and obey

$$\overline{\bigcup_{i\in\mathbb{Z}} V_i}^{L^2} = L^2(\mathbb{R}), \qquad \bigcap_{i\in\mathbb{Z}} V_i = \{0\}.$$

The response of an FIR filter is a finite sum of exponentials, hence continuously differentiable. The proof below establishes every part of the converse, including compact support and the two limiting MRA properties.

> **The two directions have different assumptions**
> Power complementarity and the gain $\sqrt{2}$ are necessary. The nonvanishing condition on $[-\pi/2, \pi/2]$ is an *additional sufficient condition* for the converse, not a necessary condition on every orthonormal scaling filter.

The theorem guarantees an MRA, not merely a formal solution of the refinement equation. This is precisely what the tent example did not guarantee.

Why the necessary conditions hold. In the orthonormal basis $\phi_{1,n}(t) = \sqrt{2}\phi(2t-n)$ of V_1, refinement gives

$$\phi(t) = \sum_n g_0[n]\phi_{1,n}(t), \qquad \phi(t-k) = \sum_n g_0[n-2k]\phi_{1,n}(t).$$

For the second identity, replace t by $t-k$ in the first and reindex the sum. Taking the inner product, conjugating the coefficients in its second argument, yields

$$\begin{aligned}\delta[k] &= \langle \phi, \phi(\cdot - k)\rangle_{L^2} \\ &= \sum_{n,m} g_0[n] g_0^*[m-2k] \langle \phi_{1,n}, \phi_{1,m}\rangle_{L^2} \\ &= \sum_n g_0[n] g_0^*[n-2k].\end{aligned}$$

To recover the power condition directly, expand the squared DTFT and keep even differences of indices:

$$|G_0(\mathrm{e}^{\mathrm{j}\omega})|^2 + |G_0(\mathrm{e}^{\mathrm{j}(\omega+\pi)})|^2 = 2\sum_k \left(\sum_n g_0[n] g_0^*[n-2k]\right) \mathrm{e}^{-2\mathrm{j}k\omega} = 2.$$

Integrating refinement also gives

$$1 = \sqrt{2}\sum_n g_0[n] \int_{\mathbb{R}} \phi(2t-n)\,\mathrm{d}t = \frac{1}{\sqrt{2}} \sum_n g_0[n],$$

which proves (11.13).

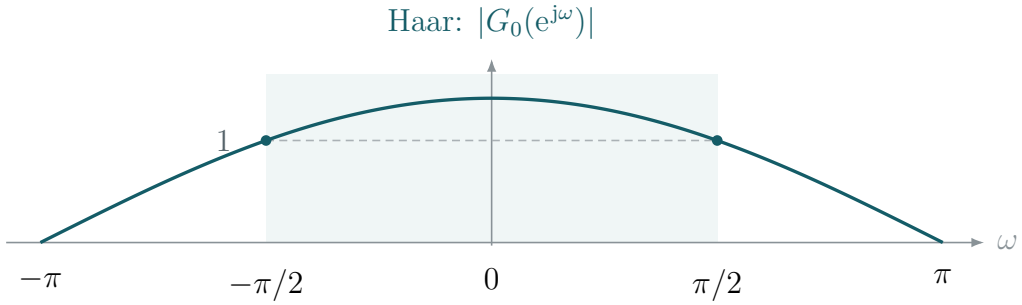


Figure 11.5: The nonvanishing interval is the shaded middle half of one frequency period. For Haar the magnitude is at least one there; zeros at $\pm\pi$ do not violate the condition.

The converse has three jobs. We must show that the product defines a finite-energy function, that its integer translates are orthonormal, and that their dilations supply every resolution. Pointwise product convergence alone does not establish these conclusions. We now prove them, then recover Haar as an explicit example.

Converse, step 1: the product exists and stays nonzero near zero. Write

$$m_0(\omega) = G_0(\mathrm{e}^{\mathrm{j}\omega})/\sqrt{2}, \qquad P_I(\omega) = \prod_{i=1}^{I} m_0(\omega/2^i), \qquad P_0 = 1.$$

The assumptions give $m_0(0) = 1$ and

$$|m_0(\omega)|^2 + |m_0(\omega+\pi)|^2 = 1, \qquad |m_0(\omega)| \le 1.$$

Because the sum defining m_0 is finite, its derivative is bounded. In particular, with $L = \frac{1}{\sqrt{2}} \sum_n |n g_0[n]|$, integration of the derivative gives $|m_0(s) - 1| \le L|s|$.

The identity $\prod_{i=1}^{I+1} a_i - \prod_{i=1}^{I} a_i = (a_{I+1} - 1) \prod_{i=1}^{I} a_i$ telescopes. Since every factor has magnitude at most one, for $R > I$,

$$|P_R(\omega) - P_I(\omega)| \le \sum_{i=I+1}^{R} |m_0(\omega/2^i) - 1| \le L|\omega| \sum_{i=I+1}^{\infty} 2^{-i} = L|\omega| 2^{-I}.$$

On every bounded frequency interval this bound tends to zero uniformly. Thus the products converge uniformly there to a continuous function

$$\Phi(\omega) \coloneqq \lim_{I\to\infty} P_I(\omega), \qquad \Phi(0) = 1, \qquad |\Phi(\omega)| \le 1.$$

Separating the first I factors from the tail gives the exact identity

$$\Phi(\omega) = P_I(\omega)\Phi(\omega/2^I).$$

We also need a positive lower bound on $[-\pi, \pi]$. Let $c = \min_{|s|\le\pi/2} |m_0(s)| > 0$ and choose R so that $L\pi 2^{-R} \le 1/2$. The same telescoping estimate, applied to the tail alone, gives

$$\left| \prod_{i=R+1}^{\infty} m_0(\omega/2^i) - 1 \right| \le L\pi 2^{-R} \le \tfrac{1}{2} \qquad (|\omega| \le \pi).$$

The first R factors have magnitude at least c, and the tail has magnitude at least $1/2$. Consequently,

$$\boxed{|\Phi(\omega)| \ge b \coloneqq \tfrac{1}{2} c^R > 0 \qquad (|\omega| \le \pi).}$$

This is where the nonvanishing hypothesis enters the proof.

Step 2: finite products with controlled energy. The product P_I is periodic and is not an $L^2(\mathbb{R})$ function. Restrict it to one expanding interval:

$$F_I(\omega) \coloneqq P_I(\omega)\mathbf{1}_{[-2^I\pi,\,2^I\pi)}(\omega), \qquad A_I[k] \coloneqq \int_{\mathbb{R}} |F_I(\omega)|^2 \mathrm{e}^{\mathrm{j}k\omega}\,\mathrm{d}\omega.$$

Here $k \in \mathbb{Z}$. We will show that $A_I[k] = 2\pi\delta[k]$ for every $I \geq 0$.

For $I \geq 1$, put $T = 2^I\pi$. Split the integral into $[-T, 0)$ and $[0, T)$, and set $v = \omega + T$ in the first part. If $i < I$, then $T/2^i = 2^{I-i}\pi$ is a multiple of 2π. Thus

$$P_{I-1}(v - T) = P_{I-1}(v), \qquad \mathrm{e}^{\mathrm{j}k(v-T)} = \mathrm{e}^{\mathrm{j}kv}.$$

The last factor has its argument shifted by π. Combining the two integrals therefore gives

$$\begin{aligned} A_I[k] &= \int_0^T |P_{I-1}(v)|^2 \mathrm{e}^{\mathrm{j}kv} \left(|m_0(v/2^I - \pi)|^2 + |m_0(v/2^I)|^2\right) \mathrm{d}v \\ &= \int_0^T |P_{I-1}(v)|^2 \mathrm{e}^{\mathrm{j}kv}\,\mathrm{d}v. \end{aligned}$$

The expression in parentheses equals one by power complementarity and periodicity. The remaining integrand has period T, so its integral over $[0, T)$ equals its integral over $[-T/2, T/2)$:

$$A_I[k] = \int_{-2^{I-1}\pi}^{2^{I-1}\pi} |P_{I-1}(v)|^2 \mathrm{e}^{\mathrm{j}kv}\,\mathrm{d}v = A_{I-1}[k].$$

At $I = 0$, the product is one. Therefore

$$A_0[k] = \int_{-\pi}^{\pi} \mathrm{e}^{\mathrm{j}k\omega}\,\mathrm{d}\omega = \begin{cases} 2\pi, & k = 0, \\ 0, & k \neq 0, \end{cases}$$

where the second case follows by integrating the exponential and using $\mathrm{e}^{\mathrm{j}k\pi} = \mathrm{e}^{-\mathrm{j}k\pi} = (-1)^k$. Induction proves

$$\boxed{A_I[k] = 2\pi\delta[k], \qquad \|F_I\|_{L^2}^2 = 2\pi.}$$

The expanding frequency window has constant total energy. We must still show that none of this energy escapes to arbitrarily high frequencies as I grows.

Step 3: convergence in energy and orthonormal translates. For any fixed $B > 0$, uniform convergence on $[-B, B]$ gives

$$\int_{-B}^{B} |\Phi(\omega)|^2 \, \mathrm{d}\omega = \lim_{I\to\infty} \int_{-B}^{B} |F_I(\omega)|^2 \, \mathrm{d}\omega \le 2\pi.$$

Letting B increase shows that $\Phi \in L^2(\mathbb{R})$ and $\|\Phi\|_{L^2}^2 \le 2\pi$. Inside the support of F_I, the identity from step 1 and its lower bound give

$$|\Phi(\omega)| = |P_I(\omega)| \, |\Phi(\omega/2^I)| \ge b|F_I(\omega)|.$$

Outside that support $F_I = 0$, so the inequality $|F_I| \le b^{-1}|\Phi|$ holds everywhere. Hence

$$|F_I - \Phi|^2 \le 2(1 + b^{-2})|\Phi|^2.$$

This bound makes the convergence in energy explicit. On $[-B, B]$, the integral of $|F_I - \Phi|^2$ tends to zero by uniform convergence. Outside this interval it is at most

$$2(1 + b^{-2}) \int_{|\omega|>B} |\Phi(\omega)|^2 \, \mathrm{d}\omega,$$

which can be made arbitrarily small independently of I. First choose B, then I. This proves $\|F_I - \Phi\|_{L^2} \to 0$, and therefore $\|\Phi\|_{L^2}^2 = 2\pi$.

Plancherel gives a function $\phi \in L^2(\mathbb{R})$ with Fourier transform Φ and $\|\phi\|_{L^2} = 1$. Explicitly, inverse Fourier transformation means

$$\frac{1}{2\pi} \int_{-B}^{B} \Phi(\omega) \mathrm{e}^{\mathrm{j}\omega t} \, \mathrm{d}\omega \xrightarrow[B\to\infty]{L^2} \phi(t).$$

Moreover, Cauchy–Schwarz gives

$$\int_{\mathbb{R}} \big| |F_I|^2 - |\Phi|^2 \big| \, \mathrm{d}\omega \le \|F_I - \Phi\|_{L^2} \big(\|F_I\|_{L^2} + \|\Phi\|_{L^2} \big) \longrightarrow 0.$$

We may therefore pass to the limit in $A_I[k]$. Using the inner product $\langle f, g \rangle_{L^2} = \int f(t) g(t)^* \, \mathrm{d}t$,

$$\langle \phi, \phi(\cdot - k) \rangle_{L^2} = \frac{1}{2\pi} \int_{\mathbb{R}} |\Phi(\omega)|^2 \mathrm{e}^{\mathrm{j}k\omega} \, \mathrm{d}\omega = \delta[k].$$

Thus the integer translates of ϕ are orthonormal.

Step 4: the box-seeded cascade converges to this function. Let $u_0 = \mathbf{1}_{[0,1)}$ and $u_{I+1} = \mathrm{C}_{g_0} u_I$. Power complementarity implies

$$\sum_n g_0[n] g_0^*[n-2k] = \delta[k].$$

Indeed, expanding the power identity gives $1 = \sum_k (\sum_n g_0[n] g_0^*[n-2k]) \mathrm{e}^{-2\mathrm{j}k\omega}$; multiplication by $\mathrm{e}^{2\mathrm{j}r\omega}$ and integration over $[-\pi, \pi]$ extracts the coefficient $\delta[r]$.

The integer translates of u_0 are orthonormal, since the unit cells are disjoint. If the translates of u_I are orthonormal, then so are $\{\sqrt{2} u_I(2 \cdot - n)\}_{n \in \mathbb{Z}}$. Expanding the next refinement gives

$$\begin{aligned} \langle u_{I+1}, u_{I+1}(\cdot - k) \rangle_{L^2} &= \sum_{n,r} g_0[n] g_0^*[r-2k] \\ &\qquad \times \langle \sqrt{2} u_I(2 \cdot - n), \sqrt{2} u_I(2 \cdot - r) \rangle_{L^2} \\ &= \sum_n g_0[n] g_0^*[n-2k] = \delta[k]. \end{aligned}$$

Induction shows $\|u_I\|_{L^2} = 1$ at every stage.

Write $U_I = \widehat{u}_I$. The cascade formula gives

$$U_I(\omega) = P_I(\omega) \widehat{u}_0(\omega / 2^I), \qquad \widehat{u}_0(s) = \int_0^1 \mathrm{e}^{-\mathrm{j}st} \, \mathrm{d}t.$$

Since $|\mathrm{e}^{-\mathrm{j}st} - 1| \le |s| t$, we have $|\widehat{u}_0(s) - 1| \le |s|/2$. Thus $U_I \to \Phi$ uniformly on bounded intervals. Both transforms have squared L^2 norm 2π.

To control their inner product, split the integral at B. On $[-B, B]$, uniform convergence gives $\int U_I \Phi^* \to \int |\Phi|^2$. The remaining part satisfies

$$\left| \int_{|\omega| > B} U_I \Phi^* \, \mathrm{d}\omega \right| \le \sqrt{2\pi} \left(\int_{|\omega| > B} |\Phi|^2 \, \mathrm{d}\omega \right)^{1/2},$$

which tends to zero as $B \to \infty$, uniformly in I. Consequently $\langle u_I, \phi \rangle_{L^2} \to 1$, and

$$\boxed{\|u_I - \phi\|_{L^2}^2 = 2 - 2 \operatorname{Re} \langle u_I, \phi \rangle_{L^2} \longrightarrow 0.}$$

This proves convergence for the specified unit-box seed, not for an arbitrary initial function.

Step 5: compact support, normalization, and nested spaces. Choose integers $a < b$ containing the support of g_0. Refinement sends a function supported in $[\alpha, \beta]$ to one supported in $[(\alpha + a)/2, (\beta + b)/2]$. Starting from $[0, 1]$, induction gives

$$\operatorname{supp} u_I \subset [a(1 - 2^{-I}),\ b + (1 - b)2^{-I}].$$

For every $\varepsilon > 0$, these supports eventually lie in $[a - \varepsilon, b + \varepsilon]$. Since $u_I \to \phi$ in L^2, the integral of $|\phi|^2$ outside that interval is zero. Letting $\varepsilon \downarrow 0$ shows $\operatorname{supp} \phi \subset [a, b]$.

All iterates and the limit lie in the fixed bounded interval $D = [\min(0, a), \max(1, b)]$. Cauchy–Schwarz gives

$$\|u_I - \phi\|_{L^1} \le |D|^{1/2} \|u_I - \phi\|_{L^2} \longrightarrow 0.$$

Their integrals are one, so $\int \phi = 1$. This also shows that the ordinary Fourier integral of ϕ equals the continuous product Φ everywhere. Continuity of C_{g_0}, proved in Section 11.2, now yields $\mathrm{C}_{g_0}\phi = \phi$.

Define $\phi_{i,n}(t) = 2^{i/2}\phi(2^i t - n)$ and $V_i = \overline{\operatorname{span}\{\phi_{i,n} : n \in \mathbb{Z}\}}^{L^2}$. Changing variables gives

$$\langle \phi_{i,n}, \phi_{i,r} \rangle_{L^2} = \langle \phi(\cdot - n), \phi(\cdot - r) \rangle_{L^2} = \delta[n - r].$$

Thus these functions are an orthonormal basis of their closed span V_i. Refinement, followed by $r = n + 2k$, gives

$$\begin{aligned} \phi_{i,k}(t) &= \sqrt{2}\, 2^{i/2} \sum_n g_0[n] \phi(2^{i+1} t - 2k - n) \\ &= \sum_r g_0[r - 2k] \phi_{i+1,r}(t). \end{aligned}$$

Every basis function of V_i is in V_{i+1}, hence $V_i \subset V_{i+1}$.

Translation by $2^{-i}k$ replaces $\phi_{i,n}$ by $\phi_{i,n+k}$, so V_i is invariant under these translations. The norm-preserving map $f(t) \mapsto \sqrt{2} f(2t)$ takes $\phi_{i,n}$ to $\phi_{i+1,n}$ and maps V_i onto V_{i+1}. These are the translation and scaling rules in the MRA definition. The two limiting properties remain to be checked.

Step 6: fine resolutions approximate every finite-energy signal. Let Π_i denote the orthogonal projection onto V_i:

$$\Pi_i f = \sum_n \langle f, \phi_{i,n}\rangle_{L^2} \phi_{i,n}, \qquad \|f - \Pi_i f\|_{L^2}^2 = \|f\|_{L^2}^2 - \sum_n |\langle f, \phi_{i,n}\rangle_{L^2}|^2.$$

This is the basis projection identity; it also gives $\|\Pi_i f\|_{L^2} \le \|f\|_{L^2}$. First suppose $\widehat{f}$ vanishes outside $[-B, B]$, and take i large enough that $2^i\pi > B$. The scaling and translation formula gives

$$\widehat{\phi}_{i,n}(\omega) = 2^{-i/2} \mathrm{e}^{-\mathrm{j}\omega n/2^i} \Phi(\omega/2^i).$$

Parseval followed by $\theta = \omega/2^i$ therefore yields

$$\begin{aligned} \langle f, \phi_{i,n}\rangle_{L^2} &= \frac{2^{-i/2}}{2\pi} \int_{\mathbb{R}} \widehat{f}(\omega) \Phi(\omega/2^i)^* \mathrm{e}^{\mathrm{j}\omega n/2^i} \,\mathrm{d}\omega \\ &= \frac{2^{i/2}}{2\pi} \int_{-\pi}^{\pi} \widehat{f}(2^i\theta) \Phi(\theta)^* \mathrm{e}^{\mathrm{j}n\theta} \,\mathrm{d}\theta. \end{aligned}$$

These are Fourier-series coefficients, up to the factor $2^{i/2}$ and a sign of the index. The Fourier-series Parseval identity is

$$\sum_n \left| \frac{1}{2\pi} \int_{-\pi}^{\pi} F(\theta) \mathrm{e}^{\mathrm{j}n\theta} \,\mathrm{d}\theta \right|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |F(\theta)|^2 \,\mathrm{d}\theta.$$

Applying it and changing variables back gives

$$\|\Pi_i f\|_{L^2}^2 = \frac{1}{2\pi} \int_{-B}^{B} |\widehat{f}(\omega)|^2 |\Phi(\omega/2^i)|^2 \,\mathrm{d}\omega.$$

Because Φ is continuous at zero with $\Phi(0) = 1$, the factor $|\Phi(\omega/2^i)|^2$ tends uniformly to one on $[-B, B]$. Thus $\|\Pi_i f\|_{L^2}^2 \to \|f\|_{L^2}^2$, so $\Pi_i f \to f$.

For general $f \in L^2$, let f_B have transform $\widehat{f}\mathbf{1}_{[-B,B]}$. Plancherel gives $\|f - f_B\|_{L^2} \to 0$. Since projection cannot increase the norm,

$$\|f - \Pi_i f\|_{L^2} \le 2\|f - f_B\|_{L^2} + \|f_B - \Pi_i f_B\|_{L^2}.$$

Choose B first and then i. Hence $\Pi_i f \to f$ for every f, proving that the union of the spaces V_i is dense in L^2.

Step 7: arbitrarily coarse resolutions retain only zero. First take f supported in $[-B, B]$. Cauchy–Schwarz on that interval gives

$$|\langle f, \phi_{i,n}\rangle_{L^2}|^2 \le \|f\|_{L^2}^2 \int_{-B}^{B} 2^i |\phi(2^i t - n)|^2 \, \mathrm{d}t = \|f\|_{L^2}^2 \int_{-2^i B - n}^{2^i B - n} |\phi(u)|^2 \, \mathrm{d}u.$$

For sufficiently negative i, these intervals are disjoint. Because ϕ is supported in $[a, b]$, at most $b - a + 3$ of them meet its support. If E_i is their union intersected with $[a, b]$, then

$$\|\Pi_i f\|_{L^2}^2 \le \|f\|_{L^2}^2 \int_{E_i} |\phi(u)|^2 \, \mathrm{d}u, \qquad |E_i| \le (b - a + 3)\, 2^{i+1} B \longrightarrow 0.$$

Why does the integral vanish? For any $T > 0$,

$$\int_{E_i} |\phi|^2 \le T|E_i| + \int_{\{|\phi|^2 > T\}} |\phi|^2.$$

The second term is arbitrarily small for large T, since $|\phi|^2$ is integrable. With T fixed, the first term tends to zero. Therefore $\Pi_i f \to 0$ as $i \to -\infty$.

For arbitrary f, truncate it in the signal domain: $f_B = f\mathbf{1}_{[-B,B]}$. Then $\|f - f_B\|_{L^2} \to 0$ and

$$\|\Pi_i f\|_{L^2} \le \|f - f_B\|_{L^2} + \|\Pi_i f_B\|_{L^2}.$$

Again choose B first and then take $i \to -\infty$. This proves the coarse-resolution limit for every f. If $f \in \bigcap_i V_i$, then $\Pi_i f = f$ at every resolution, so this limit forces $f = 0$.

> **What the proof has established**
>
> The normalized product defines a compactly supported, unit-energy scaling function. The box-seeded cascade converges to it in L^2, its integer translates are orthonormal, and the resulting nested spaces have both MRA limits. No step follows merely from drawing a convergent-looking cascade.

A normalized correspondence. The product uniquely determines an integrable refinement solution with integral one. In the other direction, orthonormal coordinates recover the unique taps $g_0[n] = \langle \phi, \sqrt{2}\phi(2 \cdot - n)\rangle_{L^2}$. Multiplying a refinement solution by a nonzero constant leaves its filter unchanged; integral one removes this amplitude freedom.

11.8 Recovering the Haar scaling function from its filter

Start with the filter, not with a guessed scaling function:

$$g_0[0] = g_0[1] = \frac{1}{\sqrt{2}}, \qquad G_0(z) = \frac{1 + z^{-1}}{\sqrt{2}}.$$

It satisfies the theorem. Its taps sum to $\sqrt{2}$, and

$$|G_0(\mathrm{e}^{\mathrm{j}\omega})|^2 = 1 + \cos\omega,$$
$$|G_0(\mathrm{e}^{\mathrm{j}(\omega+\pi)})|^2 = 1 - \cos\omega.$$

Their sum is two. Also

$$|G_0(\mathrm{e}^{\mathrm{j}\omega})| = \sqrt{2}\cos(\omega/2) \geq 1 \qquad (|\omega| \leq \pi/2).$$

Now calculate the scaling function explicitly. The normalized response is

$$m_0(\theta) = \frac{1}{2}(1 + \mathrm{e}^{-\mathrm{j}\theta}),$$

so the finite product is

$$P_I(\omega) = \frac{1}{2^I} \prod_{i=1}^{I} (1 + \mathrm{e}^{-\mathrm{j}\omega/2^i}). \tag{11.15}$$

Why this product contains every dyadic phase once. From each factor choose either 1 or its exponential. A choice $(b_1, \ldots, b_I) \in \{0, 1\}^I$ contributes

$$\mathrm{e}^{-\mathrm{j}\omega \sum_{i=1}^{I} b_i 2^{-i}} = \mathrm{e}^{-\mathrm{j}\omega k/2^I}, \qquad k = \sum_{i=1}^{I} b_i 2^{I-i}.$$

Binary representation gives every integer $k = 0, \ldots, 2^I - 1$ exactly once. Hence

$$\boxed{P_I(\omega) = \frac{1}{2^I} \sum_{k=0}^{2^I-1} \mathrm{e}^{-\mathrm{j}\omega k/2^I}.} \tag{11.16}$$

For example, with two factors, the four phases are

$$1, \quad \mathrm{e}^{-\mathrm{j}\omega/4}, \quad \mathrm{e}^{-\mathrm{j}\omega/2}, \quad \mathrm{e}^{-3\mathrm{j}\omega/4}.$$

The same dyadic product identity appeared in the staircase calculation. Here it acts on complex exponentials rather than on cell heights.

Evaluate the geometric sum before passing to the limit. Set $M = 2^I$ and $r = \mathrm{e}^{-\mathrm{j}\omega/M}$. Since

$$(1-r)\sum_{k=0}^{M-1} r^k = 1 - r^M,$$

we obtain, when $r \neq 1$,

$$P_I(\omega) = \frac{1-\mathrm{e}^{-\mathrm{j}\omega}}{M(1-\mathrm{e}^{-\mathrm{j}\omega/M})}. \tag{11.17}$$

At exceptional frequencies, the finite sum (11.16) remains the definition. For each fixed $\omega \neq 0$, the quotient is valid for all sufficiently large I.

The first-order expansion $\mathrm{e}^{-\mathrm{j}\theta} = 1 - \mathrm{j}\theta + O(\theta^2)$ gives

$$M(1-\mathrm{e}^{-\mathrm{j}\omega/M}) = \mathrm{j}\omega + O(\omega^2/M) \longrightarrow \mathrm{j}\omega.$$

Consequently,

$$\begin{aligned}\widehat{\phi}(\omega) &= \frac{1-\mathrm{e}^{-\mathrm{j}\omega}}{\mathrm{j}\omega} = \mathrm{e}^{-\mathrm{j}\omega/2}\frac{\mathrm{e}^{\mathrm{j}\omega/2}-\mathrm{e}^{-\mathrm{j}\omega/2}}{\mathrm{j}\omega} \\ &= \mathrm{e}^{-\mathrm{j}\omega/2}\frac{\sin(\omega/2)}{\omega/2}, \qquad \omega \neq 0.\end{aligned} \tag{11.18}$$

At $\omega = 0$, every factor in (11.15) equals one. Thus $\widehat{\phi}(0) = 1$, the continuous extension of (11.18).

Identify the inverse transform. Direct integration gives

$$\int_0^1 \mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t = \left[\frac{\mathrm{e}^{-\mathrm{j}\omega t}}{-\mathrm{j}\omega}\right]_{t=0}^{t=1} = \frac{1-\mathrm{e}^{-\mathrm{j}\omega}}{\mathrm{j}\omega}.$$

Therefore uniqueness of the Fourier transform yields

$$\boxed{\phi(t) = \mathbf{1}_{[0,1)}(t) \quad \text{as an } L^2 \text{ function.}} \tag{11.19}$$

Values at the two endpoints do not affect this statement. The phase factor $\mathrm{e}^{-\mathrm{j}\omega/2}$ locates the box at $[0,1]$, rather than centering it at zero.

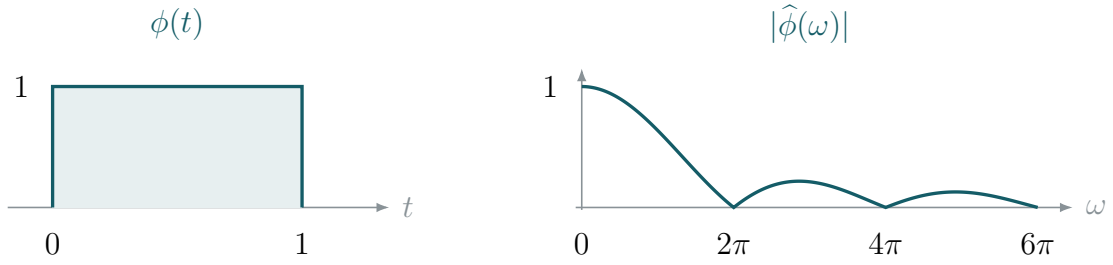


Figure 11.6: The Haar product recovers the unit box and its sinc-shaped Fourier magnitude. The right panel shows nonnegative frequencies; the magnitude is even.

11.9 Two interpretations of the Haar product

A Riemann sum. Equation (11.16) is the left-endpoint Riemann sum for

$$\int_0^1 \mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t.$$

For fixed ω, the integrand is continuous. Thus the product-to-box calculation can also be understood as an increasingly fine average of complex exponentials on $[0, 1]$.

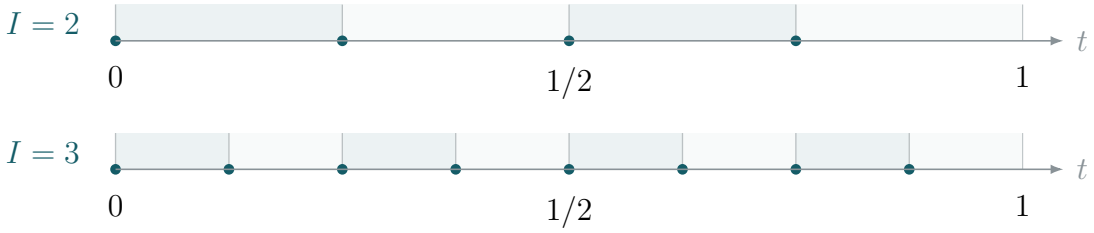


Figure 11.7: The 2^I choices in the Haar product select the left endpoints of the dyadic cells. Each term is weighted by the cell width 2^{-I}.

Euler's cosine-product identity. Factoring each term as

$$\frac{1+\mathrm{e}^{-\mathrm{j}\theta}}{2} = \mathrm{e}^{-\mathrm{j}\theta/2}\cos(\theta/2)$$

gives

$$P_I(\omega) = \mathrm{e}^{-\mathrm{j}\omega\sum_{i=1}^{I} 2^{-(i+1)}} \prod_{i=1}^{I} \cos\left(\frac{\omega}{2^{i+1}}\right).$$

The geometric sum in the exponent is $(1-2^{-I})/2$. Comparing the limit with (11.18), and writing $x = \omega/2$, yields

$$\boxed{\frac{\sin x}{x} = \prod_{i=1}^{\infty} \cos\left(\frac{x}{2^i}\right),} \tag{11.20}$$

with value one at $x = 0$. This identity also follows by iterating the double-angle formula:

$$\sin x = 2^I \sin(x/2^I) \prod_{i=1}^{I} \cos(x/2^i), \qquad 2^I \sin(x/2^I) \longrightarrow x.$$

For fixed $x \neq 0$, division is valid for all sufficiently large I; then take the limit.

The infinite product is not an arbitrary trick. It encodes successive dyadic refinements, and for Haar those refinements reproduce a familiar Fourier transform exactly.

11.10 From the scaling function to an orthogonal wavelet

Suppose g_0 is supported in $\{0, \dots, K\}$ with K odd. Form its conjugate-mirror partner and the associated function:

$$g_1[n] = (-1)^n g_0^*[K-n], \qquad \psi(t) = \sqrt{2} \sum_n g_1[n] \phi(2t-n).$$

Conjugation is needed for complex-valued taps. The same Fourier substitution used for ϕ gives

$$\boxed{\hat{\psi}(\omega) = \frac{G_1(\mathrm{e}^{\mathrm{j}\omega/2})}{\sqrt{2}} \hat{\phi}(\omega/2).} \tag{11.21}$$

The discrete orthonormal basis. Define the vectors and correlations

$$v_{\ell,k}[n] = g_\ell[n-2k], \qquad r_{\ell s}[q] = \sum_n g_\ell[n] g_s^*[n-2q].$$

The low-pass condition is $r_{00}[q] = \delta[q]$. Substitution in the mirror rule gives

$$r_{11}[q] = \sum_n g_0^*[K-n] g_0[K-n+2q] = r_{00}[q] = \delta[q],$$
$$r_{01}[q] = \sum_n (-1)^n g_0[n] g_0[K+2q-n] = 0.$$

For the last equality, pair n with $K+2q-n$: the product is unchanged, but its sign reverses because $K+2q$ is odd. Also $r_{10}[q] = r_{01}^*[-q] = 0$. Since $\langle v_{\ell,k}, v_{s,m} \rangle_{\ell^2} = r_{\ell s}[m-k]$, the family is orthonormal.

For completeness, let $e_r[n] = \delta[n-r]$ and form its finite projection $p_{r,N} = \sum_{\ell=0}^{1} \sum_{k=-N}^{N} \langle e_r, v_{\ell,k} \rangle_{\ell^2} v_{\ell,k}$. Expanding its squared error using orthonormality gives

$$0 \le \|e_r - p_{r,N}\|_{\ell^2}^2 = 1 - \sum_{\ell=0}^{1} \sum_{k=-N}^{N} |g_\ell[r-2k]|^2.$$

Thus $S_r = \sum_{\ell,k} |g_\ell[r-2k]|^2 \le 1$. The even and odd indices partition both filters, so $S_0 + S_1 = \|g_0\|_{\ell^2}^2 + \|g_1\|_{\ell^2}^2 = 2$. It follows that $S_0 = S_1 = 1$. Reindexing gives $S_{r+2} = S_r$, hence every $S_r = 1$ and $p_{r,N} \to e_r$. Every unit impulse is therefore in the closed span. Since finite truncations of any ℓ^2 sequence converge in norm, that closed span is all of $\ell^2(\mathbb{Z})$.

Transfer the coordinates to functions. The map

$$c \longmapsto \sum_n c[n]\phi_{1,n}, \qquad \phi_{1,n}(t) = \sqrt{2}\phi(2t-n),$$

preserves the norm: for finite sums, orthonormality gives squared norm $\sum_n |c[n]|^2$, and passage to the limit gives the same identity for $c \in \ell^2$. Its range is V_1. Under this map, the two discrete families become

$$\phi(t-k) = \sum_n g_0[n-2k]\phi_{1,n}(t), \qquad \psi(t-k) = \sum_n g_1[n-2k]\phi_{1,n}(t).$$

For example,

$$\begin{aligned}\langle \psi(\cdot - k), \psi(\cdot - m)\rangle_{L^2} &= \sum_{n,r} g_1[n-2k]g_1^*[r-2m]\langle \phi_{1,n}, \phi_{1,r}\rangle_{L^2} \\ &= \sum_n g_1[n-2k]g_1^*[n-2m] = \delta[k-m].\end{aligned}$$

Replacing the first g_1 by g_0 gives the cross inner product zero. The discrete completeness just proved transfers through the norm-preserving map. Hence

$$V_1 = V_0 \oplus W_0, \qquad W_0 = \overline{\operatorname{span}\{\psi(\cdot - n) : n \in \mathbb{Z}\}}^{L^2}.$$

Dilation gives $V_{i+1} = V_i \oplus W_i$, where $W_i = \overline{\operatorname{span}\{\psi_{i,n} : n \in \mathbb{Z}\}}^{L^2}$ and $\psi_{i,n}(t) = 2^{i/2}\psi(2^i t - n)$. If $r < i$, then $W_r \subset V_{r+1} \subset V_i$ while $W_i \perp V_i$; thus different detail spaces are orthogonal.

Their union is complete because the projection onto W_i is $Q_i = \Pi_{i+1} - \Pi_i$. For integers $A < B$, telescoping gives

$$\sum_{i=A}^{B-1} Q_i f = \Pi_B f - \Pi_A f.$$

The two MRA limits proved above give $\Pi_B f \to f$ as $B \to \infty$ and $\Pi_A f \to 0$ as $A \to -\infty$. Thus the sum of the detail projections converges to f in L^2. Consequently,

$$\boxed{L^2(\mathbb{R}) = \overline{\bigoplus_{i\in\mathbb{Z}} W_i}^{L^2}, \qquad \{\psi_{i,n} : i, n \in \mathbb{Z}\} \text{ is an ONB.}} \tag{11.22}$$

The direct sum means finite sums of elements of mutually orthogonal detail spaces; its closure includes their L^2 limits.

11.11 Completing the Haar example

For Haar, $K = 1$ and the conjugate-mirror convention gives

$$g_1[0] = \frac{1}{\sqrt{2}}, \qquad g_1[1] = -\frac{1}{\sqrt{2}}.$$

Substituting the recovered scaling function in its two-scale expansion,

$$\begin{aligned}\psi(t) &= \sqrt{2}\left(\frac{1}{\sqrt{2}}\phi(2t) - \frac{1}{\sqrt{2}}\phi(2t-1)\right)\\ &= \mathbf{1}_{[0,1/2)}(t) - \mathbf{1}_{[1/2,1)}(t).\end{aligned}$$

The low-pass taps add two half-width boxes; the high-pass taps subtract them.

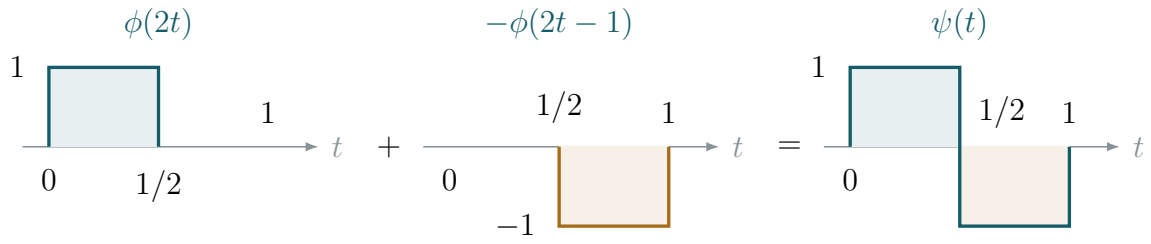


Figure 11.8: The high-pass taps convert the recovered scaling function into the Haar wavelet. The positive and negative halves have equal area, so $\int_{\mathbb{R}} \psi(t)\,\mathrm{d}t = 0$.

The sign of the partner is a convention. Replacing g_1 by $-g_1$ replaces ψ by $-\psi$. Its detail coefficients also change sign, so each synthesis term $\langle f, \psi_{i,n}\rangle_{L^2}\psi_{i,n}$ is unchanged. We use the same sign as Lecture 10.

A final check on the tent. For the three-tap filter,

$$m_0(\theta) = \frac{1}{4}(1 + \mathrm{e}^{-\mathrm{j}\theta})^2 = m_{\mathrm{Haar}}(\theta)^2.$$

Its limiting product is therefore

$$\widehat{\tau}(\omega) = \left(\frac{1 - \mathrm{e}^{-\mathrm{j}\omega}}{\mathrm{j}\omega}\right)^2 = \mathrm{e}^{-\mathrm{j}\omega}\left(\frac{\sin(\omega/2)}{\omega/2}\right)^2.$$

The value at $\omega = 0$ is one. The convolution of two unit boxes at t is the length of $[0,1] \cap [t-1, t]$: this is t for $0 \le t \le 1$, $2 - t$ for $1 \le t \le 2$, and zero outside. Thus $\tau = \mathbf{1}_{[0,1)} * \mathbf{1}_{[0,1)}$, agreeing with the cascade calculation.

A filter specifies a refinement rule, and the infinite product identifies its normalized limit. Under the fundamental theorem's hypotheses, that limit generates an orthonormal wavelet basis.